



CHEF-DR® II Pulsed Field Electrophoresis System

Instruction Manual and Applications Guide

**Catalog Numbers
170-3612
through
170-3619**

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Warranty

Bio-Rad Laboratories' **CHEF-DR** II chamber, drive module, and accessories are warranted against defects in materials and workmanship for one year. If any defects occur in the instruments during this warranty period, **Bio-Rad** Laboratories will repair or replace the defective parts free. The following defects, however, are specifically excluded:

1. Defects caused by improper operation.
2. Repair or modification done by anyone other than **Bio-Rad** Laboratories or an authorized agent.
3. Damage caused by field switcher or power supply supplied by anyone other than BioRad Laboratories.
4. Use of fittings or spare parts supplied by anyone other than **Bio-Rad** Laboratories.
5. Damage caused by accident or misuse.
6. Damage caused by disaster.
7. Corrosion caused by improper solvent or sample.

This warranty does not apply to parts listed below:

1. Fuses
2. Tubing

For any inquiry or request for repair service, contact **Bio-Rad** Laboratories after confirming the model and serial number of your instrument

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Section 1

General Information

1.1 Introduction

Pulsed field electrophoresis is a technique for resolving chromosome size DNAs. Conventional electrophoresis does not permit resolution of DNA fragments larger than 50,000 base pairs. By alternating the electric field between spatially distinct pairs of electrodes, DNAs on the order of 10 megabases are able to reorient and move differentially through the pores in an agarose gel.¹⁻³

The CHEF-DR II system (Figure 1) is an advanced pulsed field system based on the CHEF (clamped homogeneous electric fields) technique.³ The key to high resolution, sharp bands, straight lanes, and reproducible separations is a uniform electric field at all points of a gel, and an optimal 120 degree angle of alternating pulses. The CHEF-DR II system accomplishes both of these by establishing the electric field along the contour of a hexagonal array of 24 electrodes. Electrodes along the sides of the hexagon are powered in such a way as to generate two different alternating electric field vectors. Each electrode "clamps" the voltage of its individual region of space as necessary to maintain field homogeneity.



Fig. 1. The CHEF-DR II electrophoresis system.

The CHEF-DR II system features *dynamic regulation* of the individual electrode voltages. Proprietary circuitry in the drive module continually senses the voltages at individual electrodes, and actively regulates them to the proper level. Variations in gel thickness, buffer conductivity, or temperature are compensated so that uniform fields are achieved at all times, and at all points of the gel.

1.2 Specifications

The specifications below pertain to the complete system. The **CHEF-DR II** system is also available as a basic unit, which includes the gel chamber, drive module, pump, and accessories. Detailed specifications for the switcher and power supply are given in the Pulsewave 760 Switcher and Model 200/2.0 power supply instruction manuals, respectively.

CHEF-DR II Specifications

Gel chamber	
Construction	acrylic
Dimensions	11 x 43 x 44 cm
Weight	9 kg
Buffer capacity	2 l
Electrodes	platinum, 0.010 inch diameter, replaceable
Drive module	
Chassis	aluminum
Dimensions	14 x 26 x 43 cm
Weight	7.8 kg
Maximum voltage	300 V DC
Minimum voltage	25 v
Operating temperature range	0-35 °C, humidity 0-95% without condensation
Pulsewave 760 Switcher	
Maximum voltage	760 V DC
Maximum current	500 mA
Weight	5.0 kg
Power supply	
Maximum Voltage	200 V DC
Maximum current	2.0 A
Weight	5.0 kg
Pump	
Voltage	120 V AC, ground isolated, variable speed
Flow rate	1 liter/minute, typical
Weight	2 kg
Chemical compatibility	
The CHEF-DR II chamber is not compatible with chlorinated hydrocarbons (<i>eg.</i> , chloroform), aromatic hydrocarbons (<i>eg.</i> , toluene, benzene), or acetone. Use of organic solvents voids all warranties.	
Cooling recommendations	
The Model 1000 Mini Chiller (170-3654) or a refrigerated circulating water bath is recommended to keep the buffer temperature at 14 °C, for optimal results.	
Fuses	
1 A slow blow for 120/100 V 0.5 A slow blow for 220/240 V 0.5 A fast blow	
Total system weight	
28 kg	

Section 2

Description of Major Paris

2.1 Electrophoresis Chamber

The CHEF-DR II electrophoresis chamber consists of a 43 x 44 cm (17" x 17.5") acrylic box with 24 horizontal electrodes arranged in a hexagon. (See Figure 1.) Gels are electrophoresed horizontally, submerged under recirculated buffer. A 14 x 12.7 cm (5.5" x 5") gel is cast in a separate casting stand, removed, and placed in the center of the hexagon. It is held in place by two gel stops, which are inserted into the two innermost holes on the chamber floor. DNA migration is in the direction of the arrow on the lid. Other holes accommodate other gel sizes.

The individual electrodes are replaceable for easy maintenance (see Section 10). The electrode units are each sealed with an O-ring and silicone sealant to provide double protection against leakage.

There are two small chambers below the level of the main chamber floor, at the front and rear of the main chamber. These chambers allow buffer to drain and permit easy priming of the pump. Buffer enters these lower chambers through six holes at the front of the box.

There are two ports for buffer circulation. Buffer enters the main electrophoresis chamber from the rear and flows in the direction of the arrow on the lid. The plastic block at the entry point baffles the flow so the gel is not disturbed. Buffer exits the chamber at the front through the T fitting. The base of the box has four leveling screws for even gel submersion.

The lid contains two handles to allow easy lid removal, and an interlock for safety. Current from the external power supply entering the drive module passes through a short path in the lid interlock before it reaches the drive module circuitry. If the lid is removed, the current flow is broken. There is a second interlock circuit from the Pulsewave 760 Switcher to the drive module. If the Pulsewave 760 Switcher is off, there will be no current to the chamber. A third interlock requires a power supply voltage of 25 V or more for the drive module to operate.

Warning: There are high voltages and currents within the chamber, which can be harmful. Do not attempt to circumvent these safety interlocks. Always turn off the external power supply when working within the gel box.

2.2 Drive Module

The drive module is a separate electronic device which distributes and maintains the individual electrode voltages in the gel chamber. It consists of proprietary circuitry for "clamping" the individual electrode potentials so that uniform fields are obtained at all points of the gel and at all times. That is, it provides dynamic regulation of the potentials so that the proper voltages are obtained regardless of gel size, or fluctuations in buffer conductivity or temperature. If a given electrode potential begins to change due to a change in the buffer conductivity, it is automatically readjusted to the proper level.

Figure 2A shows the relative potentials of each electrode pair. When the A channel from the Pulsewave 760 Switcher is activated, the net current vector is from NW to SE. The highest potentials are found along the SE segment of the hexagon. The potentials gradually decline along the adjacent segments. The segment directly opposite the SE has zero potential, represented in the diagram as the negative terminals. Figure 2B shows the

relative potentials when the B channel is activated. The net vector is from NE to SW. The highest potential is found along the SW segment of the hexagon, with declining potentials along the adjacent segments, until there is zero potential along the NE segment, as shown in Figure 2B. The relative orientation of the electric field vectors with the A and B channels activated, constitutes the optimum 120° reorientation angle.

Note: The **CHEF-DR II** does not perform **FIGE** (Field Inversion Gel Electrophoresis). For this and other variable angle functions use the CHEF Mapper™ electrophoresis system.³²

The drive module allows a maximum of 300 volts DC from the power supply. If the incoming voltage exceeds this, the fuse will blow. The minimum operating voltage is 25 V.

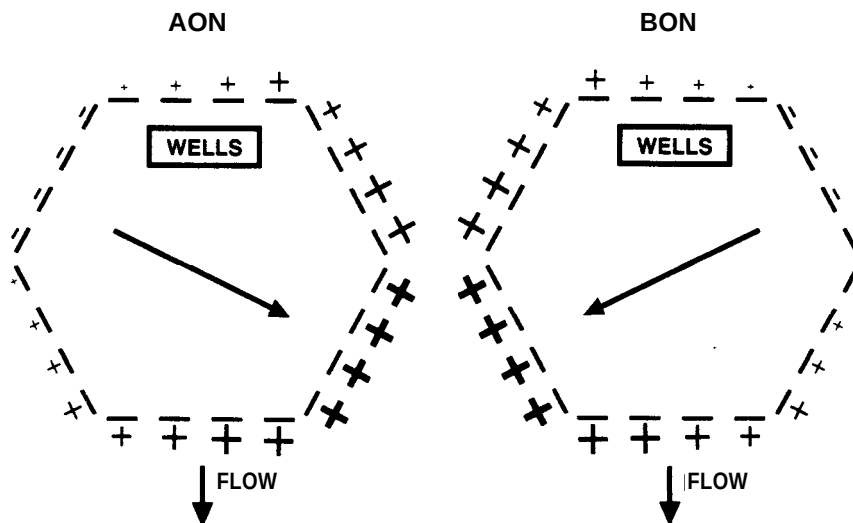


Fig. 2. Voltage clamping by the CHEF-DR II system. A. Relative electrode potentials and field vector when A channel of the Pulsewave 760 Switcher is on. **B.** Relative electrode potentials and field vector when B channel is on. The average vector during the run is in the direction of the arrow marked FLOW

The front panel of the drive module contains switches, jacks, and a fuse as shown in Figure 3.

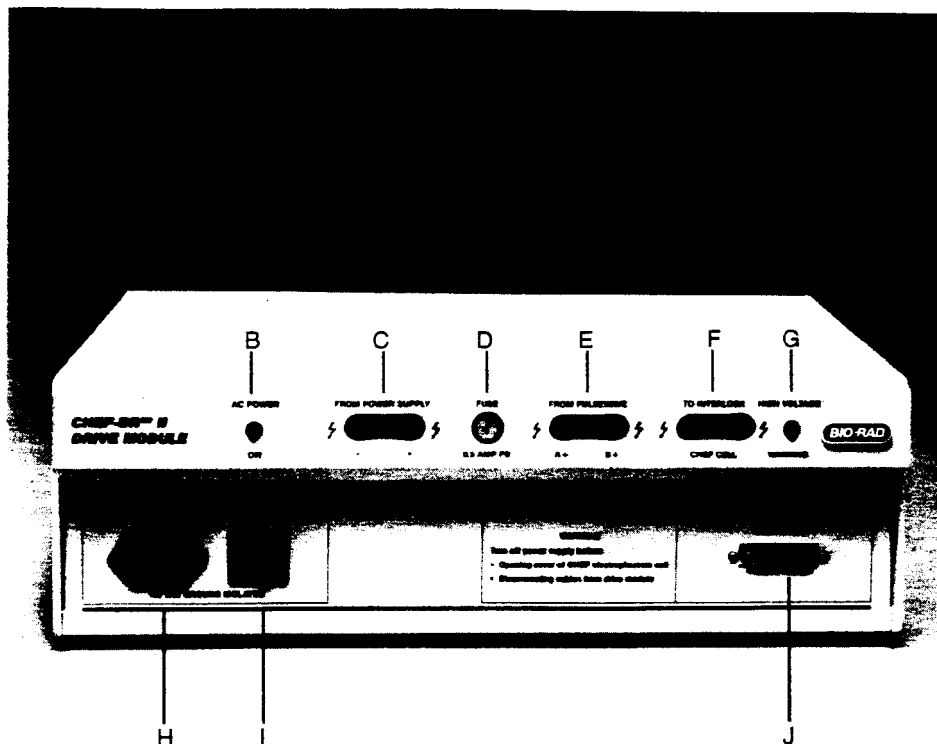


Fig. 3. CHEF-DR II drive module. **A.** Drive module power switch (rear of module). **B.** AC power indicator light; lit when module is turned on. **C.** Leads from power supply. **D.** 0.5 Amp fuse, fast blow: protects circuitry in drive module if power supply exceeds 300 V. **E.** Leads from Pulsewave 760 Switcher The A+ and B+ jacks of Pulsewave 760 Switcher are used (red leads). **F.** Interlock to CHEF-DR II electrophoresis cell. The coiled wire from the cell connects here as a safety precaution. When disconnected, the unit will not operate. **G.** High voltage indicator light; light glows in proportion to the voltage applied. THE LID TO THE CHAMBER SHOULD NOT BE OPENED WHILE THIS LIGHT IS ON. TURN OFF THE POWER SUPPLY FIRST. **H.** Power cord for pump. **I.** Pump switch. **J.** Output cable to CHEF-DR II electrophoresis cell.

2.3 Pump and Accessories

Each basic unit or system includes a variable speed pump. The pump provides a suitable flow rate of buffer through the chamber. Substitution of other pumps could pose a safety hazard and **cause** improper flow, and therefore lower resolution. The pump's power supply is electrically isolated within the drive module for safety. Its voltage requirement is independent of the line voltage supplied to the drive module (e.g. 120, 100, 220, or 240 volts). This pump should not be plugged into any equipment other than the CHEF-DR II drive module.

The pump is connected to Tygon® or plastic tubing. This tubing circulates buffer in and **out** of the chamber. The tubing may also pass through a water chiller. In this case, the pump should be located after the chiller, so that buffer flows through the chiller and then to the pump. Typically, the dial is set at 70, for about 1 l/min.

Caution: The buffer flowing in the tubing is electrically active. Care should be taken not to handle the tubing or exposed liquid while the power supply is on. Tube connections should be made with the power supply off.

A casting stand, 10 well comb, and comb holder are included for casting the gel. A 10 sample plug mold is provided for preparing **agarose** blocks. Details are given in Section 4. A sample of yeast **Chromosomal** DNA is included with catalog numbers 170-3612 through 170-3615 to help you calibrate the system. A sample of **Chromosomal** Grade **agarose** is included for megabase (> 1 mb) samples.

2.4. Pulswave 760 Switcher

The Pulswave 760 Switcher alternates the fields for the **CHEF-DR II** chamber, sets the length of the switching intervals, and sets the **electrophoresis** run time. The Pulswave 760 Switcher permits ramping the switch time from the beginning of the run to the end of the run. Special modes allow pause, countdown, linking to a computer, recovery from power failure, and automatic conversion from one set of switch conditions to another after a specified run time. These and other major features of the Pulswave 760 Switcher are described in the Pulswave 760 Switcher instruction manual.

2.5 Model 200/2.0 Power Supply

The Model 200/2.0 power supply provides up to 200 V for CHEF **electrophoresis**. If 200-300 volts are required, the Model 1000/500 power supply may be substituted. The Model 200/2.0 power supply regulates the voltage to $\pm 2\%$, sufficient for uniform fields in the CHEF chamber. Other power supplies may have up to $\pm 10\%$ fluctuation of voltage, which can lead to noticeable distortion in the fields.

For details of operation of the Model 200/2.0 power supply, see the power supply instruction manual.

Section 3

General instructions

3.1 Set-up

Place the **CHEF-DR II** chamber on a level bench surface, with the drive module to the right or on a shelf above. Place the Model 200/2.0 power supply and the Pulsewave 760 Switcher on top of the drive module.

Wiring of the system is as shown in Figure 4. Attach the power cords for the Pulsewave 760 Switcher, Model 200/2.0 power supply, and **drive** module. Be sure all units are off. Each connection is at the rear of the corresponding instrument. Plug in the pump at the left front of the drive module, at the socket labeled TO PUMP.

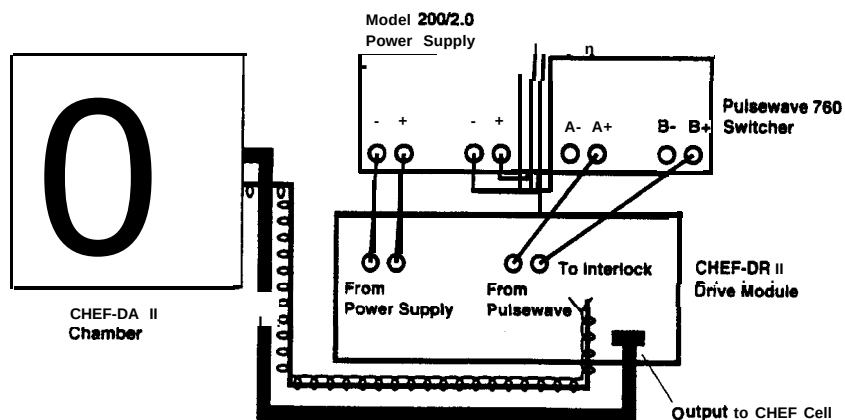


Fig. 4. Wiring the **CHEF-DR II** system.

Next, connect the leads. Connect one set of + and - (red and black) leads from the left set of jacks on the Model 200/2.0 power supply to the input jacks on the drive module labeled FROM POWER SUPPLY. This pair of leads provides power to the drive module circuitry. Connect the rightmost set of leads from the power supply to the rear of the Pulsewave 760 Switcher. This pair of leads is controlled by the Pulsewave 760 Switcher.

Connect two red leads to the A+ and B+ channels of the Pulsewave 760 Switcher. Place the other ends into the A+ and B+ jacks on the front of the drive module. Next, connect the coiled interlock cable from the **CHEF-DR II** chamber to the jacks labeled TO INTERLOCK on the drive module. Finally, connect the 25 pin cable to the port labeled OUTPUT TO CHEF CELL, and the opposite end to the port of the **CHEF-DR II** Chamber.

Connecting the Pump

If a Model 1000 Mini Chiller (170-3654) was purchased with the **CHEF-DR II** system, follow the instructions provided in the Mini Chiller manual. If a general purpose, refrigerated, water bath will be used for buffer chilling, attach 9.5 feet of 3/8" internal diameter Tygon tubing to the left outlet port of the T fitting on the **CHEF-DR II** system. This is at the front of the chamber when the arrow on the lid is pointing toward you. Connect the other end of the tubing to the inlet of the **CHEF-DR II** pump. Connect another 2 feet of tubing from the outlet of the pump to the connector at the rear of the chamber. Coil approximately 4.5 feet of the longer tubing within the bath of the water

chiller. Connect a 6" piece of tubing to the right side of the front T fitting, and clamp it off; this is used to drain the chamber after the run.

Fill the chamber with approximately 2 liters of **0.5X TBE** buffer. Later, this will be adjusted so that the gel is covered by 1 or 2 mm. It is very important that the **CHEF-DR II** cell be leveled within 1/2 degree (about 3 mm difference in buffer depth across the chamber). Use the four leveling feet to level the chamber.

Turn on the water chiller approximately 1/2 hour before the run, and set it to give 14 °C in the gel chamber. If a chiller is not used, see Section 9.3 before continuing. Begin circulating the buffer through the chiller. Turn off the pump before positioning the gel in the chamber.

3.2 Operation

This section describes general operation. See Sections 4-8 for sample preparation, gel casting, gel running, and removal.

Insert the two white gel stops into the appropriate positioning holes in the front section of the chamber's base. Place the gel in the chamber so that the lower edge rests against the two white gel stops. Remove buffer by draining until the level is 1 or 2 mm above the surface of the gel. If the gel is unevenly covered, turn the appropriate leveling screws at the corners of the **CHEF-DR II** chamber. Turn on the pump. Check the buffer temperature in the chamber with a thermometer.

Maintain a flow rate that is strong enough to keep the gel against the two gel stops, but not so strong that it causes the gel to float to the surface of the buffer. Adjust the flow rate if necessary by adjusting the dial. (Flow rate should be approximately 1 liter per minute.) Replace the lid and turn on the Pulsewave 760 Switcher and Model 200/2.0 power supply.

Important! Set the Pulsewave 760, first, then the 200/2.0 power supply.'

Set the Pulsewave 760 Switcher conditions. First set the switch interval (in seconds) by pressing INITIAL A TIME on the Pulsewave 760 panel. Raise the time to the desired interval by pressing RAISE. Set the FINAL A TIME to the same value (no ramp) or a different value (ramp). The B times are automatically calculated from the ratio, and in CHEF are always equal to the A times. Be sure the START RATIO is set to 1.0, the default value (the END RATIO defaults to the same value). Then, set the RUN TIME in hours. At this time, the unit begins switching the output of the power supply. You may pause the switching before setting the power supply, by going to Mode 0. This is done by pressing INITIAL A TIME and RUN TIME simultaneously, pressing LOWER until 0 is displayed. After the power supply is at the proper voltage, press RAISE to Mode 1.

Set the Model 200/2.0 power supply. Press VOLTS on the front panel, and raise to the desired value with RAISE. AMPS and LIMITS need not be used. The timer, if used, should be set to correspond to the RUN TIME used in setting the Pulsewave 760 Switcher. Otherwise the no load sensor of the Model 200/2.0 power supply will turn the power supply off when the RUN TIME expires on the Pulsewave 760 Switcher.

Note: not all commercial power supplies have an effective no-load sensor. With these units, the power supply should be manually shut off when the run time expires.

If all interlocks are in place and conditions set properly (including power supply voltage over 25 V), the **CHEF-DR II** system will be fully operational. Gas bubbles should begin to form at the electrodes. If not. check all cable connections, or refer to Section 11, Troubleshooting.

Section 4

Sample Preparation

4.1 Agarose Blocks

Standard procedures for DNA preparation do not yield intact, very high molecular weight DNA molecules. Large DNA molecules are so fragile that they are broken by mechanical forces during their isolation. To prevent breakage of large DNA molecules, intact cells embedded in **agarose** are **lysed** and **deproteinized** in situ. The **agarose** matrix protects the embedded DNA from shear forces and provides an easy way to manipulate samples. Processed **agarose** plug-DNA inserts are loaded directly into sample wells of **agarose electrophoresis** gels.¹⁵³

Sample inserts are cast in **Bio-Rad's** sample mold, catalog number 170-3622. The sample mold produces 20 x 9 x 1.2 mm **agarose** blocks. The block thickness allows efficient diffusion of enzymes during sample preparation and permits samples to be loaded into wells formed with **Bio-Rad's** standard well-forming combs.

Briefly, samples are prepared as follows. Intact cells are mixed at a predetermined concentration with molten Low Melt Preparative Grade **Agarose** (catalog number 162-0017). Samples are **aliquoted** into the chambers of the mold and allowed to cool, then removed from the mold and incubated with detergents and enzymes to remove all of the cellular components from the DNA. After thorough washing, **agarose** plugs are cut to appropriate size and placed in the wells of the **agarose electrophoresis** gel. A detailed procedure for preparation of *Saccharomyces cerevisiae* chromosomes is given in Section 4.3.

4.2 Liquid Samples

High molecular weight DNA can be prepared by standard procedures. DNA fragments of up to several hundred kilobases do not require preparation in **agarose** blocks, but can be added to the wells in liquid form. When running only liquid-samples, the best resolution and sharpness of bands can be achieved by using a thin well comb (0.75 mm).

4.3 Detailed Procedure for *Saccharomyces cerevisiae* Chromosomes

The following is a modified procedure from Schwartz and Cantor (1986).¹⁵³ With some modification, this procedure will work with most yeasts.

1. Grow yeast cells at 30 °C to stationary phase in **YPD** media (1% yeast extract, 2% dextrose, 2% **bactopeptone**) for 36-48 hours.
2. Pellet yeast cells by **centrifugation** at 4 °C for 10 minutes at 3,000 rpm.
3. Wash the cell pellet twice with 0.05 M **EDTA**, pH 8.0.
4. Discard **supernatant** and resuspend cells with an equal volume of 0.05 M **EDTA**, pH 8.0.
5. Determine the concentration of the yeast by counting a diluted sample on a **hemocytometer** or by plating an **aliquot** on a **YPD** plate overnight at 30 °C.
6. Add **Lyticase** [Sigma] (2 mg/ml in 0.01 M sodium phosphate containing 50% glycerol) to a final concentration of 0.5 mg/ml.
7. Place cell enzyme suspension in a 37 °C incubator for 20-30 minutes.

8. Prepare a 1.0% Low Melt Preparative Grade **Agarose** solution in 0.125 M **EDTA, pH 7.5**, by melting **agarose** in a microwave oven or on a hot plate. Let the **agarose** solution cool to 50 °C.
9. Warm the cell suspension to the same temperature as the **agarose** (50 °C). Add the 1% melted **agarose** to the cell enzyme suspension to give a final **agarose** concentration of 0.75%. Addition of 0.05 M **EDTA, pH 8.0** may be necessary at this step to achieve the desired **agarose** concentration. Stir the mixture well with a sterile, disposable transfer pipette.
10. Pipette into mold chambers and allow to cool at 4 °C for 20 minutes. Each well in the **Bio-Rad** mold chamber holds approximately 0.25 ml of sample.
11. Remove the **agarose** samples from the mold using a clean spatula, and place them into a petri dish containing LET buffer (0.45 M **EDTA pH 8.0**, 0.01 M **Tris pH 7.5**, 7.5% **β-mercaptoethanol**). For 1 liter of LET buffer, use the following recipe:

Component	Amount
0.5 M EDTA, pH 8.0	925 ml
Trizma, pH 7.5	1.5 g
β-mercaptoethanol	75 ml

12. Incubate the samples at 37 °C overnight (16-24 hours).
13. Remove the LET buffer and wash the **agarose** plug three times (15 min) in 0.05 M **EDTA, pH 8.0**.
14. Add NDS buffer to the plugs (0.01 M **Tris pH 7.5**, 0.45 M **EDTA pH 8.0**, 1% **Laurylsarcosine**, 1 mg/ml **Proteinase K**). For 1 liter of NDS buffer, use the following recipe:

Component	Amount
0.5 M EDTA, pH 8.0	890 ml
Trizma, pH 7.5	1.5 g
N-Laurylsarcosine	100 ml
100 mg/ml Proteinase K	10 ml

15. Incubate overnight at 50 °C (16-24 hours).
16. Remove the NDS buffer and wash the **agarose** plugs with 0.05 M **EDTA, pH 8.0**, for 15 minutes at room temperature.
17. Remove the wash solution and wash again with 0.05 M **EDTA, pH 8.0**. Repeat this step again.
18. Do a final wash by adding 0.05 M **EDTA, pH 8.0** and store at room temperature overnight.
19. **Agarose** plugs can be stored for at least a year in 0.05 M **EDTA, pH 8.0**, at 4 °C.

Section 5

Casting the Gel

Casting the gel requires the following components: a casting stand with removable end plates; a comb; and a comb holder. The casting stand which comes with the **CHEF-DR II** is 14 cm x 12.7 cm (5.5" x 5"). Other sizes are available: 21 cm x 14 cm (8.5" x 5"), and 14 cm x 21 cm (5.5" x 8.5").

1. Position the two end plates over the screws from the top, with the screw slots pointing down (see Figure 5). Tighten the screws finger tight. The endplates are removed to allow hardened gel to be slid into the **electrophoresis** chamber. Attach a comb to the comb holder and adjust the height of the comb to 2 mm minimum above the casting stand surface. Place the comb holder so that the comb is 5-10 mm from one of the end plates.
2. Prepare the desired concentration of **Bio-Rad's agarose** in an **electrophoresis** buffer of choice (TBE or TAE, see below). Melt the **agarose** in a microwave oven or on a hot plate.

Compositions of Buffer Stock Concentrates:

10X TBE Buffer:

108 g Tris base
55 g boric acid
40 ml 0.5 M EDTA, pH 8.0
water to 1000 ml

50XTAE Buffer:

242 g Tris Base
57.1 ml glacial acetic acid
100 ml 0.5 M EDTA, pH 8.0
water to 1000 ml

3. The gel should be cast on a level surface and **Bio-Rad's Leveling Table**, catalog number 170-4046, is recommended for this purpose. For 5 mm thick gels, pour 80-100 ml of molten **agarose** into the standard casting stand and allow it to cool for 1 hour at room temperature. Carefully remove the comb holder and comb; it is sometimes helpful to rock the holder back and forth slightly during its removal. Sample plugs can be added to the wells while the gel remains in the casting stand.

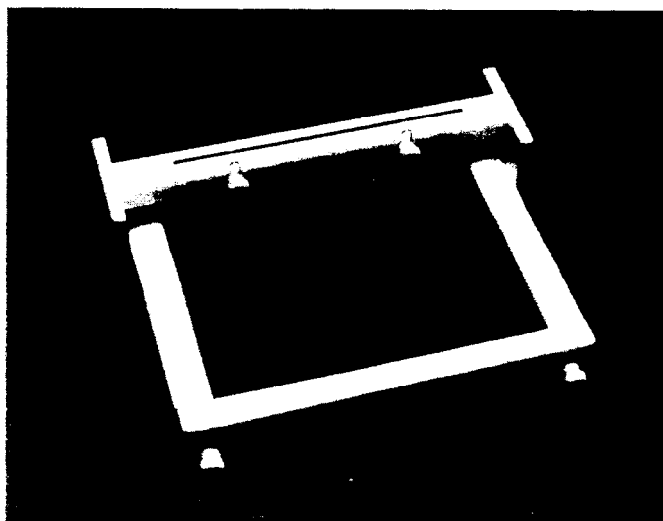


Fig. 5. The CHEF-M II casting stand and comb holder.

Section 6

Loading the Samples

One of the following methods should be used to load the sample.

1. DNA in a sample plug should be placed on a smooth clean surface, and cut to size using a razor blade or spatula. Samples should be less than **90%** of the height of the wells. Place **agarose** plugs onto the front walls of the sample wells using a spatula and gently press them into the bottoms of the wells. Press the plugs firmly against the front walls of the wells. Fill each sample well with Low Melt Preparative Grade **Agarose** at an **agarose** concentration equal to that of the gel and allow the **agarose** to harden at room temperature for **10 to 15** minutes.
2. Alternatively, the sample plug can be cut into blocks and placed on each tooth of the comb. Cast around the comb. The plug will remain in place when the comb is removed.
3. Liquid samples can be added to the sample wells with the gel positioned under the **electrophoresis** buffer in the chamber. Turn the pump off when adding liquid samples to prevent samples from washing out of the wells. Run the samples into the gel for approximately **5 to 10** minutes before turning the pump back on.

Section 7

Running the Gel

7.1 Running the Gel

Level the **CHEF-DR II electrophoresis** chamber by adjusting the leveling screws at the comers. Place gel stops in the two innermost holes in the lower (downstream) portion of the chamber. Pour 2 liters of buffer (appropriate concentration of **TBE** or **TAE**) into the chamber. Turn on the variable speed pump, and allow the buffer to equilibrate to desired temperature (we recommend **14 °C** buffer temperature in the chamber) with the pump tubing coiled in a temperature controlled water bath. Remove the endplates from both ends of the casting stand and slide the gel onto the surface of the chamber under the buffer. Place the gel in the chamber so that the bottom rests against the two gel stops. Check the buffer level to ensure that the gel is covered by about 2 mm buffer. Maintain the maximum flow rate without disturbing the gel (approximately 1 liter per minute). Adjust the buffer flow if necessary by using the flow adjuster knob on the variable speed pump. Turn on the Pulsewave **760** Switcher and power supply (refer to Pulsewave **760** manual for complete operating instructions). Set Pulsewave **760** parameters and voltage (see Section **9**, Applications).

Note: Using gel stops at the top of the gel will result in distortion of the outer lanes.

Section 7.2 DNA Size Standards

An example of *Saccharomyces cerevisiae* chromosomes is shown in Figure 6C.

Bio-Rad recommends running standards in each gel to allow the sizes of unknown samples to be determined and to verify the **electrophoresis** conditions. Figure 6 shows three Bio-Rad standards for pulsed field **electrophoresis**. These come as blocks of 0.75% Low-Melt agarose. Recommended running conditions are given in the figure legend.

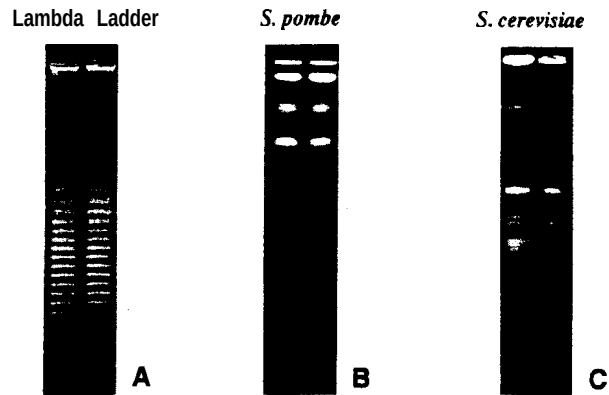


Fig. 6. A. Lambda Ladder (170-3635) Switch time was ramped from 50 to 90 seconds over the 22 hour run time. The gel was 1.0% agarose in 0.5X TBE. Electrophoresis was at 6 V/cm at 14 °C.
B. Schizosaccharomyces pombe (170-3633) Strain 972 h-. Switch time was 30 minutes for 72 hours. The gel was 0.6% Chromosomal Grade agarose in 0.5X TAE. Electrophoresis was at 1.5 V/cm at 14 °C.
C. Saccharomyces cerevisiae (170-3605) Strain YNN295. Switch time was ramped from 60 seconds to 120 seconds over 24 hour run time. The gel was 1.0% agarose in 0.5X TBE. Electrophoresis was at 6 V/cm at 14 °C.

Section 8 Removing and Staining the Gel

Turn off your power supply and the Pulsewave 760 **Switcher**. Place gel into 0.5 µg/ml **ethidium** bromide solution and let stain for 20-30 minutes. (**Caution: Ethidium** bromide is a mutagen. Always wear gloves while handling gels or solutions containing the dye.) Destain in distilled H₂O for 1-3 hours. The DNA can be visualized by placing the gel on a **UV transilluminator** (254-360 nm). Remove the buffer from the gel box by unclamping the drain tube and allowing the buffer to drain into a 2 liter container with the pump turned off. Discard buffer and **reclamp** the drain tube. Leave the lid of the gel chamber open on one end when not in use.

Note: Leaving **electrophoresis** buffer in the gel chamber with the lid on, for prolonged periods, can result in war-page of the lid.

Section 9 Applications

9.1 Strategies for Electrophoretic Separations

There are several parameters that must be considered before performing an **electrophoretic** separation of very high molecular **weight** DNA. The separations of large DNA molecules in **agarose** gels are affected by **agarose** concentration, buffer concentration, temperature, switch times, voltage and total **electrophoresis** run time.

Agarose Concentration

The **agarose** concentration determines the size range of DNA molecules separated, and the sharpness, or tightness, of the bands. **Agarose** concentrations of 1% are useful in separating DNA molecules up to 3 **mb** in size. **Agarose** concentrations in the range of 1.2-1.5% are typically used for improved band tightness, however, run times will increase proportionately. Gel concentrations below 1% (0.5-0.9%) are useful in separations of extremely high molecular weight DNA, greater than 3 **mb**, though the bands are a bit more diffuse.

There are several **agarose** types that allow easy handling of low concentration gels. These **agaroses**, in the concentration range of 0.5-0.8%, can be used to decrease the run time for separation of large DNA (>1 **mb**). An example of this type of **agarose** is **Bio-Rad's Chromosomal Grade Agarose** (162-0133).

Buffer Concentration and Temperature

In pulsed field **electrophoresis**, the mobility of the DNA is sensitive to changes in buffer temperature. As buffer temperature increases, the mobility of the DNA increases, but the band sharpness and resolution decrease. It is recommended that the buffer be chilled to 14 °C to maintain band sharpness and to dissipate heat generated during prolonged runs. Also, buffer recirculation is required to prevent temperature gradients from occurring. High voltage runs (300 V) exceeding 1 day require buffer changes after each 24 hour period, to prevent any possible buffer degradation.

Standard **Tris Borate** or **TBE**, at a concentration of 0.5X, is the most commonly used buffer in pulsed field **electrophoresis**. **Tris-acetate** buffer, or **TAE**, at a concentration of 1.0X, can be used in place of **TBE**. Other buffer concentrations used are in the range of 0.25X-1.0X.

Switch Times

The migration rate of DNA molecules through an **agarose** gel is dependent on pulse time, voltage (field strength), pulse angle and run time. In pulsed field **electrophoresis**, DNA molecules are subjected to alternating electric fields imposed for a period of time called the switch time. Each time the field is switched, the DNA molecules must change direction or reorient in the gel matrix. Larger molecules take longer to reorient and have less time to move during each pulse, so they migrate slower than smaller molecules. Resolution will be optimal for DNA molecules with reorientation times comparable to the pulse time. So, as the DNA size increases, the pulse time needs to increase to resolve the molecules. Under some conditions, larger molecules sometimes run ahead of smaller ones.^{50 †}

†Note: Rapid switch times will cause rapid wear and eventual burnout of the electrodes. For best results, replace standard gauge (0.01") electrodes with thick gauge (0.02") electrodes. The two types may be mixed in the same chamber.

Voltage (Field Strength)

DNA migration will increase with increases in voltage or field strength. However, greater migration is accompanied by decreased band sharpness. In general, as the size of the DNA molecules increases, the field strength should decrease. At high field strengths (6 V/cm) some very large DNA (>3 mb) cannot be resolved on the gel and the field strength needs to be reduced. Moreover, some large DNA molecules will not enter the gels at high field strengths. Therefore, in selecting the field strength for an experiment, a compromise between run time and resolution has to be made.

Electrophoresis Run Time

The **electrophoresis** run time is determined by the migration rates of the DNA molecules under investigation. The migration rates, in turn, are affected by the switch time, field strength, and pulse angle. As the migration rate of the DNA molecules decreases, the **electrophoresis** run time needs to increase to adequately resolve the DNA molecules of interest.

Pulsed Field Conditions by DNA Size

In the table below run parameters are suggested that can be used for the various DNA size ranges listed.

	DNA 1-100 kb	DNA 0.1-2.0 mb	DNA 2-4 mb	DNA >4 mb
%Agarose	1.0-1.2%	0.8-1.2%	0.6-1.0%	0.5-0.8%
Buffer	0.5X TBE	0.5X TBE	1.0X TAE	1.0X TAE
Temperature	14 °C	14 °C	14 °C	14 °C
Voltage	200-300 v	150-200 v	50-100 v	40-75 v
Pulse Parameters	0.05-10 set	10-200 set	200-1800 set	10-60 min
Run Times	2-15 hr	15-30 hr	24-72 hr	72-170 hr

9.2 Megabase DNA Size Standards

The complete **CHEF-DR II** system includes a sample of yeast **chromosomal** DNA embedded in Low Melting Temperature **Agarose**. Using a scalpel, cut a small rectangular piece from the block so that it will fit in one of the wells. Follow the directions for **electrophoresis** of yeast chromosomes in Section 7, Figure 6C, when first using the **CHEF-DR II** chamber, to be sure all parameters and conditions are optimized.

S. pombe, lambda ladder, and 5kb ladder DNA size standards are also available from Bio-Rad Laboratories.

9.3 Separations at Room Temperature

Electrophoresis may be conducted at room temperature, without a chiller, but the buffer should not be allowed to exceed 30 °C. It is important to maintain the temperature at a **steady** value. To facilitate heat transfer, it is recommended that 4-5 feet of the **Tygon** tubing be coiled into a bucket of water. Alternatively, gels may be run in a cold room. Recirculation of the buffer is required. The buffer should be changed every 24 hours.

Since heat generation is proportional to the square of the voltage, it is essential to lower the voltage to 150 V or less, depending on the size of DNA to be resolved. This voltage drop is required even if gels are **electrophoresed** in a cold room. *Saccharomyces cerevisiae* should be **electrophoresed** at 125-150 V. Gel strength and buffer concentration do not need to be changed, although switch times and run times may be increased 10 to 20%. The conditions for resolution of *Saccharomyces cerevisiae* DNA are the same as those given in Section 9.1, except that the voltage is 150 V and the temperature is 29 °C.

Alternatively, the ionic strength of the buffer may be decreased to 0.25X TBE. In this case, the voltage must be decreased even more than above, or DNA will not enter the gel. In some cases, DNA bands may be more diffuse at room temperature than those resolved at 14 °C.

9.4 Casting Gels of Varying Size, Multiple Gels

Large gel formats are possible using the **CHEF-DR II** chamber. These are accomplished by positioning the gel stops in other holes near the lower center of the gel chamber.

Gels are cast using alternative size casting stands from **Bio-Rad**. An 8.25" wide x 5" long casting stand accessory allows **electrophoresis** of large numbers of samples. A 5.5" wide x 8.25" long gel assembly is available for samples with a wide range of DNA fragment sizes.

Two gels may be **electrophoresed** under identical conditions from a single Pulsewave 760 Switcher and Model 200/2.0 Power Supply. Two **CHEF-DR II** basic units are required (i.e. two chambers and two drive modules), the Extender Block accessory (catalog number 170-3604) for the Pulsewave 760 Switcher, and the Extender Cables (catalog number 170-3630). Consult **Bio-Rad** for details.

Alternatively, two gels may be stacked on top of each other. This technique allows doubling of the number of samples, or removal of the top gel to check the progress of very long runs. Some increase in lane distortion around the edges is typical, however. Consult Nagy and Chou, *Nucleic Acids. Res* **18:5317** (1990) for details.

9.5 Procedure for Mammalian DNA*

Introduction

An increasingly important application of pulsed field **electrophoresis** is the analysis of human and other mammalian DNA. With this technique, fragments which are hundreds or thousands of kilobases in length can be separated. Such fragments may contain entire gene families. Since mammalian chromosomes are too large to enter gels with current technology, they must be digested with restriction enzymes and visualized by probe hybridization. This section describes procedures for preparing samples, restriction digestion, setting switch time parameters, and using appropriate controls. The DNA employed is from human white cells. Additional information on sample preparation and pulsed field **electrophoresis** conditions may be found in the references.

Sample Preparation

The following procedure will give a final concentration of 107 cells/ml in the agarose. Typical agarose slices contain 50-100 µl, giving 5-10 µg DNA per lane.

1. Harvest cells in phosphate-buffered saline:

A. Tissue culture cells are collected by scraping or **trypsination**, followed by **centrifugation**. Suspension cultures are centrifuged directly, although brief **trypsin** digestion (0.25% in saline) can improve lysis.

B. White blood cells are prepared from whole blood by either isotonic lysis or **centrifugation** through **Ficoll** (e.g., Sigma **Histopaque**®). Starting volume is 5-50 ml blood.

C. Tissue should be minced with scissors or a razor blade and dispersed to single cells with a loose fitting **Dounce** homogenizer. Let large clumps of tissue settle, transfer **supernatant**, and centrifuge.

* Contributed by Dr. Bruce Birren, California Institute of Technology

2. Wash cells twice with PBS at 4 °C.
3. Count cells, and resuspend in PBS at a concentration of 2×10^7 cells/ml. Cells may be counted in a **hemocytometer** (e.g., **Reichert Inc.**).
4. Gently pipet up and down to break up clumps, and warm to 37 °C.
5. Mix with an equal volume of 1% low melting **agarose** prepared in PBS and cooled to 45 °C. **Note:** the **agarose** should be qualified for restriction digestion.
6. Transfer with a **Pipetman®** or disposable pipet to plug mold former, avoiding bubbles.
7. Allow samples to harden at 4 °C or on ice for 10 min.
8. Transfer sample blocks to 3-5 volumes of 0.5 M EDTA, pH 9, 1% Sarcosyl, 0.5 mg/ml Proteinase K (**Boehringer-Mannheim**).
9. Digest for 1-2 days at 50 °C with constant, gentle shaking. Samples may be stored in this solution at 4 °C.

Restriction Digestion of DNA in Agarose

1. Rinse samples twice in distilled water. Wash with 50x volume of **TE** either twice for 3 hours at room temperature or overnight at 4 °C by gently shaking or rotating in 15 or 50 ml conical bottom polypropylene tubes.
2. **Optional:** Residual **Proteinase K** may be inactivated by washing sample in **TE** with 1 mM **PMSF** (**phenyl methyl sulfonyl flouride**) twice for 2 hours at room temperature prior to final washes in **TE**. Step 1 above is usually sufficient.
3. Samples are digested with a restriction enzyme in 100 µl to 200 µl final volume, depending on the thickness of the plug. To the sample, add 10-20 µl 10x digestion buffer as recommended by the supplier of enzyme, **nuclease** free **BSA** to 0.1 mg/ml, distilled water, and enzyme.
4. Digest **agarose** blocks with 2-10 U enzyme per µg DNA. Incubate 4 hours to overnight at the appropriate temperature. Enzymes may be added twice, once at the beginning and again partway through the incubation. This is recommended for those enzymes that have short lifetimes. (Refer to the manufacturer's specifications.)
5. Stop the reaction by adding 1 ml 0.5 M EDTA. Incubate 5 minutes at room temperature.
6. Remove EDTA from sample. Place **agarose** slices into the wells of an **agarose** gel. It is important that the blocks are flat against the front wall of the well; remove any trapped bubbles. Fill the void space with 0.8% low melting **agarose**. Alternatively, place blocks on each tooth of the gel forming comb, and cast gel around the comb.
7. Place gel in chamber, and equilibrate at the running temperature for 0.5 hour prior to starting the run.

Electrophoresis and Results

The use of pulsed field electrophoresis to identify large fragments of human DNA homologous to a **cDNA** probe is illustrated in Figure 7. Human DNA was digested with two different rare cutting enzymes and separated for 36 hours prior to blotting and hybridization. With **Mlu** I, three bands are visible between about 450 and 600 kb. **Nru** I yields two bands in this size range, one at 450 and one at 670 kb. In addition, with **Nru** I there is a faint hybridization signal from the region near the top of the gel that contains unresolved large DNA. The yeast markers indicate that the upper range of resolution of

this gel is **900 kb**. Thus, a hybridization signal in this upper region could indicate partial digestion products or the presence of fragments above **2 mb**. In the stained gel, Figure **8A**, only the largest *S. cerevisiae* chromosomes are resolved, though the *S. pombe* chromosomes are well separated. When the **Nru** I digest is run under the same conditions, the two bands which had been well resolved in Figure 7 are visible only as a smear. However, the two larger bands are now distinguishable, one at about **2 mb** and the other at **2.5 mb**. Not I yields a single large band at about **2.2 mb**.

The importance of using the appropriate **switch** interval when analyzing pulsed field data is further apparent from Figure 9. An **Mlu** I digest of human DNA was run using three different pulsed field conditions prior to blotting and hybridization. Each of the gels resolved a different size range of **DNA**s, as indicated by the marker sizes in the figure. Under conditions where only fragments from **200 to 600 kb** were resolved, four bands were visible on the **autoradiogram**. When longer switch intervals were employed, so that fragments up to **1,600 kb** are separated, three bands are visible. In this experiment the central band appears darker than the other two. Often, however, they are of uniform intensity and provide no clue to the doublet lurking there. Using conditions which resolve bands above **2 mb**, only a smear is seen on the **autoradiogram**, in a position corresponding to the massed *S. cerevisiae* chromosomes. Thus, to obtain accuracy in both sizing and in establishing the number of bands in Southern blots, gel conditions that maximize resolution in the area containing the bands of interest must be **used**.

Numerous other factors can influence the ability to detect and accurately size bands of interest in pulsed field gels. The most common reasons for failure include:

1. The DNA samples are degraded by **nuclease** either during the sample preparation or during the restriction digestion. Degradation during preparation will produce plugs which yield small DNA when run directly after preparation. Each new batch of samples should be tested for contaminating **nuclease** by performing a mock digestion and running the products on a pulsed field gel. The mock digestion consists of **incubation** at the same temperature and time in a restriction digestion reaction minus the restriction enzyme (*i.e.*, DNA + buffer + **BSA**). Samples which are free of **nuclease** will have almost all the DNA remaining in the plug or moving with the limiting mobility region of large unresolved DNA.
2. Incorrect gel conditions are used. Size markers should indicate that the conditions used (especially voltage and switch time) were appropriate for resolving the size region of interest. Each new digest must be analyzed under a variety of gel conditions. Generally, it is best to start with the conditions which give as broad a range of sizes as possible to give a rough estimate of the number and sizes of bands (this is often facilitated by using a switch time ramp). Then gels which focus on a narrower size range can be **used** to more precisely determine the sizes of the fragments.
3. Incomplete restriction digestion. This can result from protein remaining in the samples or from incomplete removal of the agents used to prepare the DNA (**EDTA**, **protease**, detergent). In addition, because many of the enzymes used to generate large fragments are sensitive to **methylation**, it can be extremely difficult to cut particular **methyated** sites. Using DNA prepared from different sources (e.g., **fibroblasts** instead of blood) can often give different length fragments due to tissue-specific **methylation** patterns.

4. Sample overloading. The apparent size of fragments in pulsed field gels is strongly dependent on the amount of DNA loaded in the gel. With increasing amounts of DNA, bands are retarded in their migration, indicated larger than true sizes. Figure 10 shows hybridization to two different amounts of the same restriction digest run side by side. Two distinct bands are visible in lane 1. As expected, an increased amount of DNA leads to broadening of both bands. Furthermore, the midpoint of each of the broad bands is shifted higher in the gel. This is particularly clear for the band near the bottom of the **autoradiogram**. A lighter exposure of the pattern in lane 2 indicates a larger size for each of the two bands than is seen in lane 1.

Conclusion

In these experiments a cDNA probe was used to localize those large restriction fragments bearing homology to the sequence of interest. Multiple bands were detected with each of the enzymes used. Numerous situations can give rise to multiple bands on blots with large DNA. These include the presence of multiple regions homologous to the probe (both genes and pseudogenes), partial digestion due to restriction enzyme inhibitors in the samples or interference with cutting by **methylation**, or **allelic** differences between chromosomes that make up the sample. Distinguishing between each of these cases requires both knowledge of the factors which influence pulsed field mobility and application of standard molecular biological methods to explain the nature of each of the bands.

References

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 Lai, E., Birren, B. W., Clark, S. M., Simon, M. I., and Hood, L., *BioTechniques*, **7**, 34-42 (1989).
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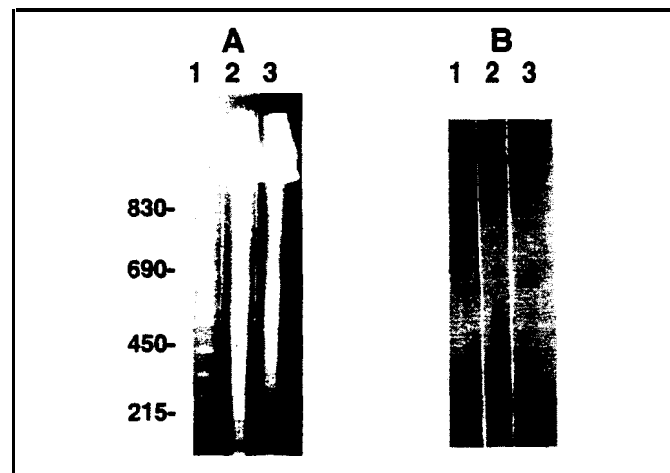


Fig. 7. Separation and identification of fragments smaller than 1 megabase. **A.** Gel stained with ethidium bromide. Human DNA in agarose blocks (5 µg/lane) was digested for 6 hr with 40 U of enzyme in a reaction volume of 150 µl. Following digestion, samples were run in a 1% agarose gel, 6 V/cm for 36 hr in 0.5x TBE at 14 °C Switch time was a ramp from 55 set to 75 sec. Samples were: Lane 1, Yeast size standard (YNN295). Lane 2, Mlu I digest. Lane 3, Nru I digest. Sizes (in kb) are indicated to the left of the yeast standards. **B.** Autoradiogram showing the position of bands identified by the probe. After staining, the gel was exposed for 1 minute on a short wave UV light-box (254 nm, 2 mW/cm²). DNA was denatured in NaOH/NaCl, neutralized with Tris/NaCl and transferred overnight with 10x SSC to Zeta-Probe® membrane according to the manufacturer's protocol. The hybridization probe was a cDNA clone for a human G protein b subunit labeled by random priming.

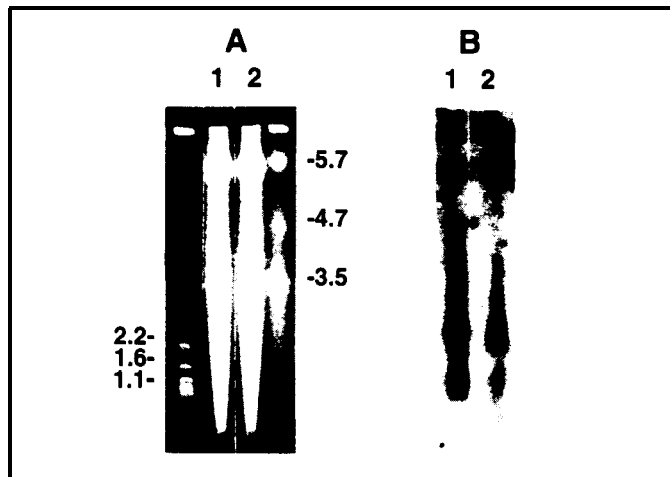


Fig. 8. Separation and identification of fragments larger than 1 megabase. The samples were separated in 0.8% agarose for 108 hr in 1x TAE at 14 °C, 2 V/cm. Switch time was 35 min. **A.** Ethidium bromide stained gel. *S. cerevisiae* chromosomes are on the far left, *S. pombe* on the right, with sizes indicated in megabases. Lane 1, Nru I; Lane 2, Not I. **B.** Autoradiogram after hybridization. Hybridization was as described In Figure 7.

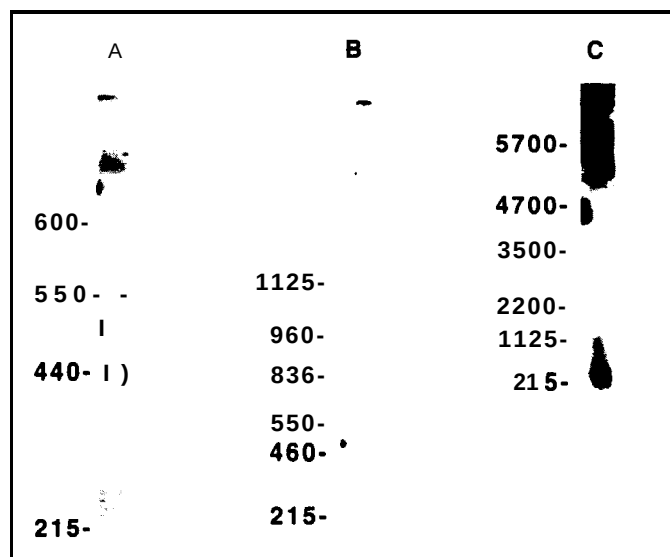


Fig. 9. Importance of switch interval for resolving fragments. Human DNA was digested with Mlu I for 6 hr and separated with different switch times or other pulsed field parameters. After membrane separation, DNA was transferred to Zeta-Probe membrane as described In the legend for Figure 7. **A.** 1% gel electrophoresed for 36 hr at 6 V/cm with a switch ramp from 30 set to 60 sec. **B.** 1% gel electrophoresed 36 hr at 6 V/cm with a ramp from 95 set to 115 set **C.** 0.8% gel electrophoresed 108 hr at 2 V/cm with a switch time of 35 min. Sizes indicate the position of the yeast markers included in each gel.

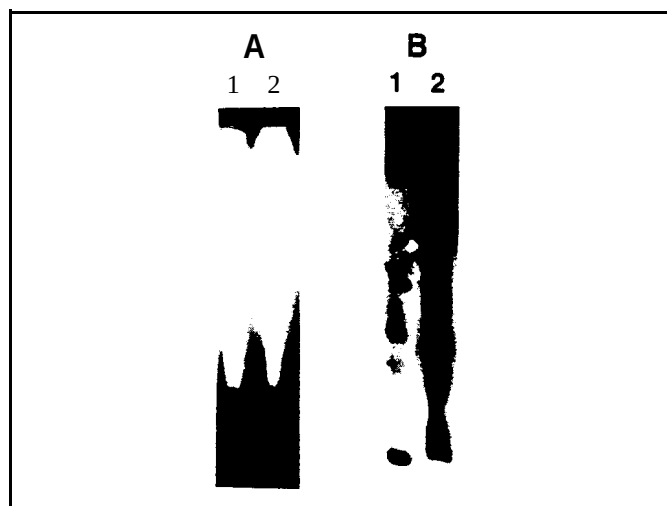


Fig. 10. Effects of sample overloading on hybridization signals. Human DNA was prepared in agarose simultaneously using two different cell concentrations. A sample block of identical thickness from each was digested with Not I and separated as described in the legend for Figure 9. The amounts of DNA loaded per lane were: Lane 1, 3 μ g. Lane 2, 15 μ g. **A.** Ethidium bromide stained gel, **B.** Autoradiogram after hybridization Blotting and hybridization were as described in the legend for Figure 7.

9.6 Blotting Megabase DNA's[†]

Southern Blot Transfer

Pulsed field electrophoresis has become a powerful technique for physical mapping of genes in various organisms. In order to determine the **chromosomal** location of a gene in a microorganism or the size of the restriction fragment containing a gene in mammalian systems, large DNA fragments separated by CHEF are transferred onto membranes and detected by Southern hybridization analysis. The procedures described for Southern transfer of DNA from standard agarose gels onto membranes are applicable to large DNA fragments separated by CHEF, with the addition of the gel pretreatment step listed below.

Gel Pretreatment

Since DNA fragments larger than 20 kb cannot be transferred efficiently, DNA fragments separated by pulsed field gels must be cleaved before transfer onto membranes. DNA can be cleaved by using either acid (**depurination**) or UV irradiation. The **depurination** reaction is harder to control and is extremely sensitive to temperature. Exposure to shortwave UV light is a reliable method for nicking DNA in pulsed field gels before transfer.

Procedure

The following procedure was developed for use with the **GS Gene Linker**™ UV chamber. For optimal results, this protocol must be followed rigorously.

1. -Stain the gel with 1.0 μ g/ml ethidium bromide (**EtBr**) for exactly 30 minutes with constant agitation. Use a fresh dilution of the **EtBr** stock for each gel. Do not destain the gel prior to nicking.

[†] Contributed by Dr. Eric Lai
University of North Carolina

2. Immediately **UV** irradiate the gel, using the **GS** Gene Linker chamber, with **60 mJoules** of energy. The gel should be photographed using very short exposures (cl second) to minimize exposure to **UV** radiation. The gel can also be destained if desired. Transfer the nicked DNA to nylon membrane using alkali or neutral conditions (see discussion).
3. Soak the gel in **0.4 N NaOH, 1.5 M NaCl** for **15** minutes. Transfer the DNA onto Zeta-Probe **GT** nylon membrane (162-0196) using 2 liters of **0.4 N NaOH, 1.5 M NaCl** as the transfer solvent.
4. Set up the capillary transfer as follows, from bottom to top:
 - A. Coming Pyrex glass dish (**28 x 18 x 4** cm).
 - B. A plexiglass or plastic box for support, about 3 cm high and small enough to fit in the glass dish (e.g., **Eppendorf** yellow pipette tip rack).
 - C. Glass plate (**16 x 20** cm).
 - D. Two sheets of blotting paper as a wick (**18 x 30** cm; S&S, **GB002**).
 - E. **Agarose** gel (top side down).
 - F. Zeta-Probe **GT** membrane cut to the same size as the gel and **prewetted** with distilled water.
 - G. Two sheets of blotting paper (**18 x 15** cm; S&S, **GB002**).
 - H. A stack of paper towels 10 cm high.
5. Transfer the DNA **24-48** hours.
6. Carefully remove the paper towel and blotting papers. Remove the membrane together with the gel, turn over the membrane and gel, lay them gel side up, and mark the location of the wells and the orientation marker on the top of the gel. The position of the wells can be accurately marked on the membrane by using a fine point permanent marker pen, cutting through the bottoms of the wells.
7. Neutralize the membrane in **0.5 M Tris, pH 7.0** (neutralization buffer) for 5 minutes, followed by rinsing briefly in **2x SSC**. Transferred DNA can be visualized on the membrane by placing the damp blot on a **transilluminator**.
8. Dry the membrane by blotting onto **3MM** or other adsorbent paper and proceed with hybridization. **UV** crosslinking of the DNA to the membrane is not recommended with this alkaline transfer method.

Discussion

1. The procedure is based on gels approximately 6 mm thick. If thicker gels are used, the staining period may be prolonged to allow diffusion of **EtBr** into the middle of the gels. DNA that is not stained with **EtBr** will not be nicked by the **UV** light and thus will not be transferred from the gel.
2. If the output of the **UV** light source is not known and no **UV** meter is available, you can titrate your **UV** light source as follows. Run a CHEF gel with eight lanes of *S. cerevisiae* chromosomes as markers using a switch time that will provide resolution from **200-1,000 kb**. Stain the gel with **EtBr**, and photograph with medium-wave **302 nm UV** light and fast film (Polaroid type **667**) to minimize nicking of DNA. Note the exposure time of the photo. Cut the gel into eight strips, each containing a lane of separated yeast chromosomes. **UV** irradiate the strips with a **254 nm** light source for time intervals of **5, 10, 15, 30, 45, 60, 90, and 300** seconds. If a **254 nm** light source is not available, **302 nm** light can be used, but exposure times have to be lengthened **approximately** five-fold. Alkaline transfer the gel strips as described, and stain the gels after transfer. Take a photograph of the gel strips using the same **UV** light source, film, and

exposure time as before transfer, and compare it with the photograph before transfer. Choose the time period that results in **80-90%** transfer of DNA. Do not choose the time intervals with complete transfer because most of the transferred DNA fragments will be too short for effective hybridization. If less than **10** second short-wave **UV** irradiation is required, you may need to use a **302 nm** light source for taking the picture of the gel and cutting away excess gel area. As a general rule, **10** seconds or less exposure time is needed with a new **UV transilluminator**. The **UV** output will decrease with time, to as little as **30%** of its initial rating after **7** years.

3. Presoaking the gel in **NaOH** prior to transfer decreases background and increases transfer efficiency.
4. Pulsed field gels can also be blotted onto membranes using **20x SSC** as the transfer buffer solvent with standard alkaline **denaturation** followed by neutralization. Alkaline transfer onto nylon membranes gives as good or better sensitivity as standard transfers onto **nitrocellulose** filters. The alkaline procedure is much simpler and faster. In addition, nylon membranes can be reused many more times than **nitrocellulose** filters. Some blots may be reused as many as **20** times.
5. **DNA**s separated on the **CHEF-DR II** or **CHEF Mapper** system can also be vacuum transferred onto nylon membranes in **4** hours using a commercial vacuum blotter, such as the **Model 785 Vacuum Blotter (165-5000)**, and **NaOH** as buffer.
6. The DNA is transferred from the back of the gel (the side opposite the wells) onto the membrane because irregularities in the surface of the gel frequently occur during solidification of these high percentage gels (**1%**). These surface artifacts will interfere with the transfer of the **DNA**s from the gel. Transfer from the other side of the gel insures smooth surface contact between the gel and the membrane.
7. It is essential to neutralize the membrane after transfer to prevent changing the **pH** of the hybridization buffer during hybridization.
8. It is not absolutely necessary to bake nylon membranes after alkaline transfer since the DNA should be fixed onto the membrane by **NaOH**.
9. To monitor the efficiency of the transfer, stain the gel in neutralization buffer for **30** minutes with **1 @ml EtBr**. Take a photograph of the post-transferred gel, and compare with the original picture.

9.7 Other Applications

All separations performed at 14 °C except *Dictyostelium*. Gel sizes were 5.5" x 5" except for the 8 kb - 2.2 Mb separation, which was 5.5" x 8.25" long. Pulse angle was 120° except as noted in *S. pombe*.

DNA	Size range W	Agarose cont.	Pulse Time (seconds)	Run Tie (hours)	Voltage	Buffer
Restriction fragments	0.2-23	1.2%	0.1	5	200	0.5 x TAE
5 kb Ladder	5-75	1.0%	1-6 ^a	11	200	0.5 x TBE
YACs	50-600	1.0%	20-50	26	200	0.5 x TBE
Bacterial	50-400	1.0%	30-60	22	200	0.5 x TBE
Human lymphocytes	50-1,500 50-650	1% 1%	50-90	20	200	0.5 x TBE
			25 ^b	17	200	0.5 x TBE
			50	17	200	
Restriction fragments	200-400	1.0%	24-36	36	200	0.5 x TBE
<i>Leishmania</i>	50-1,000	1.0%	35-120	45	200	0.5 x TBE
<i>S. cerevisiae</i>	200-2,200	1.0%	60-120	24	200	0.5 x TBE
Mammalian	1-50	1.0%	0.1-2.5	8	200	0.5 x TBE
Mammalian	5-150	1.0%	0.2-13	15	200	0.5 x TBE
Mammalian	5-250	1.0%	0.2-22	15	200	0.5 x TBE
Mammalian	50-200	1.0%	6.8-17.3	32	200	0.5 x TBE
<i>Mycoplasma</i>	400-3,500	1.0%	1,800-600 ^b	120	60	0.5 x TBE
			600-96	50	60	
<i>Candida albicans</i>	1,000-5,000	0.8%	120	24	150	0.5 x TBE
			240	36	150	
<i>Aspergillus</i>	1,600-5,600	0.7%	900-3,000	156	50	0.5 x TBE
<i>S. pombe</i>	3,500-5,700 106 ^c	0.6% ^c 0.8% ^c	1,800	72	50	0.5 x TAE
			1,800	32	50	0.5 x TAE
<i>Dictyostelium</i>	3,600-9,000	0.8%	2,000-7,000 ^d	158	62	0.3 x TBE
			7,000-9600	82	50	
Wide range	8-2,200	1%	60-120	20	200	0.5 x TBE
			0.9	6	200	

(a) Ramped pulse from 1 to 6 seconds over 5 hours. On the Pulsethrow 760 switcher, Initial A = 1 sec. Final A = 6 sec. B pulse is determined by the ratio.

(b) Two stage (block) procedure using modes 10 and 11 of the Pulsethrow 760 switcher. Program the first set of conditions, then press mode keys and raise to 10. Enter the next row of conditions. Finally, press mode keys and raise to 11 to &in switching.

(c) Chromosomal Grade Agarose

(d) Two stage procedure. Since there is a voltage change in the second stage, lower the voltage on the Model 200/2.0 Power supply after 158 hr. Run temperature 10 °C. See Cox et. al., *Proc Natl. Acad. Sci. USA*, **87**, 8247-8251 (1990). On the CHEF Mapper™ system, use multistate mode with two blocks. The voltage gradient in the first block is 1.9 V/cm, and 1.5 V/cm in the second block.

Section 10

Maintenance of Equipment

10.1 Replacing Electrodes

The gel chamber requires little maintenance except for rinsing after every run. Dirt and other build-up can be removed with laboratory detergent and a fine cloth. Care should be taken not to touch the electrodes.

If one of the electrodes should break, or leak at the O-ring, it may be replaced. Additional electrodes are available from **Bio-Rad** Laboratories.

To replace the electrode, turn the gel chamber upside down and remove the six screws. Lift off the base plate. Remove the hexagonal nut on the wire, then remove the nut on the electrode to be replaced, and push down firmly on the post to remove the old electrode. Turn the box over, insert the new electrode, pack with self leveling silicone sealant (available at most hardware stores), and replace the nut. Replace the wire and base plate.

If one of the pins to the serial cable bends, use tweezers to carefully straighten it. Replacement cables may be ordered from **Bio-Rad**.

10.2 Replacing a Fuse in the Drive Module

If the DC voltage exceeds 300 V entering the drive module, the 0.5 ampere **FB** (fast blow) fuse will blow. The right light on the drive module will go off. Replace the fuse by unscrewing the cartridge at the front of the drive module. Replace with 0.5 ampere **FB** fuse. Make sure the external power supply is off when replacing a fuse.

If there is a power surge, the **SB** fuse will blow. The left light on the drive module will go off. The fuse is located at the rear of the drive module. Replace the fuse with a 1.0 A **SB** if your voltage is 120 V, a 1 S A **SB** if 220 V or 240 V.

If the unit still does not operate, contact **Bio-Rad** Laboratories. Do not attempt to open and repair the drive module, or the warranty may be voided.

10.3 Maintenance of the Chamber

When the chamber is not in use, especially for extended periods, all buffer should be removed to prevent damage to the plastics.

Section 1 I

Troubleshooting Guide

Equipment:

<u>Problem:</u>	<u>Solution:</u>
No power	Check fuse at back of module
No voltage across electrodes with AC power light lit on drive module	1. Check that lid is on 2. Check that serial cable is fully inserted at both ends 3. Check that coiled interlock cable is firmly attached 4. Confirm that all leads are properly attached from the Pulsewave 760 Switcher and Model 200/2.0 power supply 5. Replace fuse in drive module 6. Confirm that external power supply is set to at least 25 V 7. Check round, 1 Amp FB fuse on rear of Pulsewave 760 Switcher 8. Check that interlocks in lid are tight
Gel floats away	Pump flow rate is too high or too low ; adjust with dial on variable speed pump
No or low buffer flow	Look for kink in tubing
Gel bands appear very distorted: lanes very curved,* sharp, but slanting	1. Foreign object in chamber (remove thermometer, gel stops at top of gel, etc. 2. Insufficient or non-uniform cooling due to low pumpflow 3. Check that buffer is level with surface of gel: use leveling feet 4. Pulsewave 760 Switcher not set with A/B ratio of 1 .0, or if other source, poor regulation of switch times 5. Replaced damaged electrode 6. Fault in drive module, contact Bio-Rad

*Slight distortion of the outermost lane is normal

Applications:

Problem:	Solution:
Bands smeary or fuzzy	1. Excessive heating: lower the voltage or ionic strength of buffer 2. Improper switch interval: see Section 9.1 3. Gel percentage too low: increase percentage 4. Sample degraded: impure enzymes, or wash cycles too short (agarose blocks) 5. Agarose impurities: consult Bio-Rad 6. Sample overload: adjust sample concentration
Large DNAs not resolved	1. Lower the voltage to below 75 V 2. Increase switch time 3. Agarose impurities
High background in lanes	1. Insufficient washing of samples 2. Sample may be contaminated with RNA or other material 3. DNA concentration too high
Distorted bands	1. Sample contains too high salt or detergent concentration 2. Buffer breakdown. Change every 24 hours 3. Wells were distorted. Recast gel 4. Sample plugs were crushed when placed in wells 5. Pump flow rate too low: check for kink along tubing
Thick bands	Use thinner wells

Section 12

References

Applications in Pulsed Field Electrophoresis

The following are references in pulsed field electrophoresis, primarily from 1987-1989. The list surveys a wide area of applications and organisms, but is not exhaustive. Underlined references use the CHEF-DR® II pulsed field electrophoresis system.

Organism	Reference numbers
<i>Aspergillus</i>	20
Bacteria	8, 49, 55, 80, 90, 93, 120, 128, 149, 152, 190, 195
<i>Candida albicans</i>	92, 110, 111, 112, 150, 170
<i>Caulobacter</i>	52
<i>Dictyostelium</i>	36
<i>Drosophila</i>	64, 204
Epstein-Barr virus	75
<i>Giardia</i>	2
<i>Histoplasma</i>	176
Human	9, 14, 27, 33, 37, 43, 50, 60, 66, 67, 82, 83, 88, 102, 122, 132, 133, 136, 137, 142, 144, 147, 183, 186, 189, 202
<i>Leishmania</i>	61, 69, 103, 135, 156, 163
Mouse	18, 19, 29, 89, 97, 179, 207
<i>Mycoplasma</i>	6, 25, 114, 139
<i>Neurospora</i>	<u>96</u> , 130
<i>Paramecium</i>	68
Plants	72, 187
<i>Plasmodium</i>	10, 42, 86, 138, 161, 171
<i>Pseudomonas</i>	7
<i>S. cerevisiae</i>	3, 23, 31, 54, 87, 91, 123, 124, 151, 162, 166, 182, 194, 205, 206
<i>S. pombe</i>	28, 53, 73, 75, 119, 169, 184
<i>Tetrahymena</i>	38
<i>Trypanosoma</i>	113, 192
<i>Ureaplasma</i>	34
Application	Reference numbers
Alkaline blotting (Zeta-Probe@ membrane)	120, 131
Blotting	1, 9, 26, 31, 66, 72, 120, 125, 130, 145, <u>202</u>
Chromosome rearrangements	127, 141, 147, 166
CHEF	13, 20, 26, 28, 30, 31, 36, 37, <u>39</u> , 42, 45, 56, <u>57</u> , 58, 61, 85, 91, 93, 94, <u>96</u> , 107, 121, 123, 136, 139, <u>143</u> , 145, 151, 156, 162, 163, 168, 171, 173, 197, 198, 201, <u>202</u>
Circular DNA	8, 75, 78, 79, 104, 118, 159, 160, 163, 164, 190, 207
Cosmid mapping	14, 46, 70, 98, 173, 193
Diagnostics (e.g. cancer)	148
DNA over 5 megabases	20, 72, 130, 197
DNA under 200,000 bases	12, 34, <u>39</u> , 77, 93
Epstein-Barr virus	75
FIGE	7, 12, 14, 15, 17, 24, 25, 33, 47, 65, 68, 77, 146, 173, 182, 187
FIGE with <i>S. pombe</i>	184

Application	Reference numbers (cont.)
Gene amplification	108, 135, 191
Mapping	1, 18, 59, 66, 87, 134, 140, 199
Minute chromosomes	26, 113
OFAGE	23, 52, 54, 79, 82, 87, 147
On-off pulsing	98, 181
PACE	11, 32, 98
PFGE	16, 50, 102, 137, 138, 140, 153, 155, 189, 184, 202
PFG in acrylamide	84
PHOGE	5
Preparative	45
Ramps	17, 107, 142
Restriction enzymes	105, 120
Review of PFGE	99
RFGE	3, 4, 9, 122, 123, 157, 159, 180
RFLP polymorphism	7, 14, 40, 43, 50, 65, 85, 102, 109, 150
Sample preparation-general	145, 155, 167, 188
Sample preparation-cell lines	167
Sample preparation-bacteria	58, 167, 172, 195
Sample preparation-YACs	45
Sample preparation-tissues	106, 167
Secondary structure analysis	30, 118, 128
Single strand DNA	181
Size standards	39, 74, 116, 200
Strain characterization	35, 69, 71, 121
TAFE	51, 62, 63, 132, 148, 179
Theory	15, 22, 44, 48, 76, 81, 100, 101, 117, 126, 129, 130, 158, 174, 175, 177, 178, 185, 196
Two-D PFG	10, 203
Virus	15, 145
Visualization of DNA (microscope)	154, 168
Yeast artificial chromosomes (YACs)	10, 16, 41, 45, 93, 94, 95, 107, 115, 144, 183, 198

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Section 13

Product Information

Catalog Number	Description
170-3670	CHEF Mapper XA Chiller System, 120 V , includes CHEF Mapper XA power module with embedded auto algorithm for protocol optimization, interactive algorithm program disc, bar code reader, electrophoresis cell, Model 1000 Mini Chiller, Variable Speed Pump, temperature probe, 12 feet Tygon® tubing, 14 cm wide x 12.7 cm long casting stand, 10 well comb and comb holder, 10 sample plug mold, leveling bubble, cables, <i>S. cerevisiae</i> standards, 5 grams Chromosomal Grade A&rose, instruction manual
170-3671	CHEF Mapper XA Chiller System, 100 V
170-3672	CHEF Mapper XA Chiller System, 220 V
170-3673	CHEF Mapper XA Chiller System, 240 V
170-3686	Algorithm Upgrade Kit , includes EPROM with embedded algorithm to convert CHEF mapper to XA, Interactive algorithm disc, bar code reader, cable: PC to CHEF Mapper
170-3612	Complete CHEF-DR II System, 120 V , includes gel box, drive module, Pulswave 760 Switcher, Model 200/2.0 power supply, pump, 5.5" x 5" casting stand and comb holder, 10 well comb and 10 sample plug mold, 12' Tygon tubing, chromosomal DNA size standard.
170-3613	Complete CHEF-DR II System, 100 V
170-3614	Complete CHEF-DR II System, 220 V
170-3615	Complete CHEF-DR II System, 240 V
170-3616	CHEF-DR II Basic Unit, 120 V , includes gel box, drive module, pump, 5.5" x 5" casting stand and comb holder, 10 well comb and 10 sample plug mold, 12' Tygon tubing.
170-3617	CHEF-DR II Basic Unit, 100 V
170-3618	CHEF-DR II Basic Unit, 220 V
170-3619	CHEF-DR II Basic Unit, 240 V
170-3600	Pulswave 760 Switcher, 120 V
170-3601	Pulswave 760 Switcher, 100 V
170-3602	Pulswave 760 Switcher, 220 V
170-3603	Pulswave 760 Switcher, 240 V
165-4761	Model 200/2.0 Constant Voltage Power Supply, 100/120 V
165-4762	Model 200/2.0 Constant Voltage Power Supply, 220/240 V
165-4710	Model 1000/500 Power Supply, 100/120 V
165-4711	Model 1000/500 Power Supply, 220/240 V

Accessories and Reagents

Catalog Number	Description
170-3654	Model 1000 Mini Chiller , 120 V
170-3688	Model 1000 Mini Chiller , 100 V
170-3655	Model 1000 Mini Chiller , 220/240 V
170-3668	External. Temperature Probe
170-3644	Variable Speed Pump , 120 V
170-4046	Leveling Table
170-3625	Gel Stops (4)
170-3643	Gel Scoop
170-3646	Standard Electrodes (0.01") , 6
170-3648	Thick Electrodes (0.02") , 6
165-5031	GS Gene Linker , UV crosslinking chamber, 120 V
165-5034	GS Gene Linker , UV crosslinking chamber, 100 V
165-5032	GS Gene Linker , UV crosslinking chamber, 220 V
165-5033	GS Gene Linker , UV crosslinking chamber, 240 V
170-3604	Pulsewave 760 Extender Block
170-3630	Extender Cables
170-3620	Casting Stand , 5.5" x 5"
170-4326	10 Well Comb , 5.5" wide, 1.5 mm thick
170-4325	10 Well Comb , 5.5" wide, 0.75 mm thick
170-4324	15 Well Comb , 5.5" wide, 1.5 mm thick
170-4323	15 Well Comb , 5.5" wide, 0.75 mm thick
170-4322	20 Well Comb , 5.5" wide, 1.5 mm thick
170-4321	20 Well Comb , 5.5" wide, 0.75 mm thick
170-4344	30 Well Comb , 5.5" wide, 1.5 mm thick
170-3626	Casting Stand , 8.25" x 5"
170-3664	Casting Stand , 14 cm wide x 21 cm long, and frame
170-3627	15 Well Comb , 8.25" wide, 1.5 mm thick
170-3628	30 Well Comb , 8.25" wide, 1.5 mm thick
170-3645	45 Well Comb , 8.25" wide, 1.5 mm thick
170-3629	Comb Holder , 8.25"
170-3622	10 Sample Plug Mold
170-3623	Prep Comb , 2 well (plus 2 outer wells for size standards), 5.5" wide, 1.5 mm thick
170-3605	DNA Size Standards , <i>S. cerevisiae</i>
170-3633	DNA Size Standards , <i>S. pombe</i>
170-3635	DNA Size Standards , lambda ladder

Catalog Number	Description
170-3624	DNA Size Standards, 5 kb ladder
162-0017	Low Melt Preparative Grade Agarose , 25 g
162-0135	Chromosomal Grade Agarose , 25 g
162-0133	Molecular Biology Certified Agarose , 100 g
162-0159	Zeta-Probe Roll, 33 cm x 3 m
162-0115	Nitrocellulose Roll , 33 cm x 3 m

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